

Enhancing Agronomic Efficiency: Investigating the Water Retention Properties of Coir/Kenaf Blended Non-Woven Mats

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Abstract:

The use of natural fibers as sustainable materials is gaining popularity due to their biodegradability. This study delves into the potential of two natural fibers, namely coir and kenaf, to be used in needle-punched nonwoven samples for various applications. The study involves developing nonwoven samples with varying blend ratios of coir and kenaf fibers, followed by testing them for specific properties such as porosity and water retention capacity. The experiment reveals that a blend of 25% coir and 75% kenaf fibers exhibits noteworthy water retention capacity, with a recorded 97.2% water-holding capacity due to its small pore size as compared to other samples. These results suggest that coir and kenaf fibers can be blended to produce nonwoven samples with desirable properties for various applications.

Keywords: biodegradability, coir, non-woven, needle punching, porosity, water retention

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1. Introduction

The agricultural sector faces issues such as climate change, water shortage, and modernization, making textiles an appropriate answer. Textiles, with their enormous surface area, lightweight, and superior water retention qualities, provide eco-friendly and biocompatible options. Agrotech is a key branch of technical textiles that uses textile structures to regulate environmental conditions. Natural biodegradable textile fibers derived from agricultural plants are widely utilized in ropes, carpet backings, and purses. The research project intends to design and manufacture biodegradable non-woven mats made from coir/kenaf and coir/recycled cotton materials for agricultural water irrigation. This seeks to provide environmentally friendly alternatives to traditional synthetic carpets. A thorough evaluation will be carried out to confirm the mats' efficacy, practicality, and overall quality.

2. Objective

To produce a non-woven mat using coir/kenaf blends with water retention properties for reducing the scarcity of water in agriculture. To reduce the water usage in the irrigation process.

3. Literature

The agricultural sector faces many threats due to climate change, water scarcity, and modernization. Water scarcity and management are serious problems, and textile is one of the suitable sectors that can offer a better solution to this. Natural biodegradable textile fibers like coir, kenaf, and cotton are eco-friendly and can be used in agriculture. Needle-punched nonwoven samples were developed with coir/cotton and coir/kenaf fibers, which have water-retention properties. Kenaf mats processed using needle-punching technology have shown potential mechanical and physical attributes for diverse applications. Coir is a biodegradable organic fibre that can retain moisture for a longer period and can be used for various agricultural applications [1, 2 & 5]. The article talks about the need to replace synthetic fibers with natural fibers in Agro-textile products, and how coir is a suitable alternative due to its moisture retention, high wet strength, and ability to improve soil fertility. The usage of organic coir mulch sheet in agricultural lands increases the quality and yield of crops in a faster way. The objective of the work was to develop an environmentally friendly method for the effective utilization of coir fibre by adopting steam pre-treatment, which increases the cellulose content in the fibre and removes lignin and hemicelluloses. Steam explosion has been proved to be a green method to expand the application areas of coir fibre. The processing of kenaf bast fibre, a renewable, biodegradable, and ecologically beneficial fibre bundle, was investigated in this work. It was discovered that

although decreasing fibre bundle strength, modified chemical degumming and alkaline sulphide treatments enhanced tensile properties, fineness, length, and softness [6-8]. Research was conducted in Tunisia to develop seed production practices of high-yielding kenaf cultivars. Kenaf is a renewable bast fibre that is environmentally safe and biodegradable. Water is retained by various particles in different ways. Larger particles are less able to retain moisture than smaller ones over extended periods of time. This is because water is easily able to escape through the larger pores in the larger particles, whereas the smaller holes in the smaller particles hold water. The number and size of a particle's holes as well as its surface area determine how much water it can hold. Because their pores better retain moisture, smaller particles absorb water more readily than bigger ones. Thus, the coir pith grades' holes are crucial for retaining water in them [3, 4 & 9].

3.1 Conclusion from the Literature Review

This text investigates the uses and qualities of natural fibers, particularly coir and kenaf, in agriculture, textiles, and environmental sustainability. These fibers, which are biodegradable and moisture-retaining, are employed in a variety of applications. Their characteristics are enhanced through processing procedures such as needle-punching, steam pre-treatment, and chemical retting. They have environmental benefits like as renewability and low impact as compared to synthetics. They also have useful qualities, such as preventing fungal.

4. Materials and Methods

4.1 Fibres

Coir and Kenaf fibres are used as a raw material in this research work. Coir is a long staple fibre and kenaf fibre is cut up to 40mm for the manufacturing process. Non-woven mat was produced on 28th march, 2024 using needle punching method and testing were done on 30th march, 2024.

The production of a non-woven mat involves the utilization of two fibers, namely coir and kenaf. Coir is a natural fibre extracted from the husk of a coconut, while kenaf is a fibre obtained from the bast of the plant. These fibers are carefully blended to create a strong, durable, and eco-friendly material that is widely used in various industries such as automotive, construction, and agriculture.

4.1.1 Coir

Coir, also called coconut fibre, is a natural fibre extracted from the outer husk of coconut, and used in products such as floor mats, doormats, brushes, and mattresses. Coir is the fibrous material found between the hard, internal shell and the outer coat of a coconut. Other uses of brown coir (made from ripe coconut) are in upholstery padding, sacking and horticulture.

Coir must not be confused with coir pith, which is the powdery and spongy material resulting from the processing of the coir fibre. Coir fibre is locally named 'coprah' in some countries, adding to confusion. Pith is chemically like coir, but contains much shorter fibers.

4.1.2 Structure of Coir

Coir fibres are found between the hard, internal shell and the outer coat of a coconut. The individual fibre cells are narrow and hollow, with thick walls made of cellulose. They are pale when immature, but later become hardened and yellowed as a layer of lignin is deposited on their walls. Each cell is about 1 mm (0.04 in) long and 10 to 20 μm (0.0004 to 0.0008 in) in diameter. Fibres are typically 10 to 30 centimetres (4 to 12 in) long. The two varieties of coir are brown and white. Brown coir harvested from fully ripened coconuts is thick, strong and has high abrasion resistance. It is typically used in mats, brushes, and sacking. Mature brown coir fibres contain more lignin and less cellulose than fibres such as flax and cotton, so are stronger but less flexible. White coir fibres harvested from coconuts before they are ripe are white or light brown in colour and are smoother and finer, but also weaker. They are generally spun to make yarn used in mats or rope.

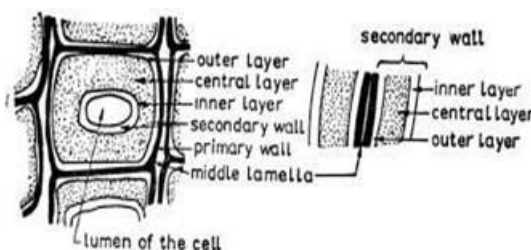


Figure 1: Structure of Coir

4.1.3 Kenaf

Kenaf, *Hibiscus cannabinus*, is a plant in the family Malvaceae also called Deccan hemp and Java jute. *Hibiscus cannabinus* is in the genus *Hibiscus* and is native to Africa, though its exact origin is unknown. The name also applies to the fibre obtained from this plant. Kenaf is one of the allied fibres of jute and shows similar characteristics.

Kenaf is cultivated of its fibre in India, United States of America, Indonesia, Malaysia, South Africa, Viet Nam, Thailand, parts of Africa, and to a small extent in southeast Europe. The stems produce two types of fibre: a coarser fibre in the outer layer (bast fibre), and a finer fibre in the core. The bast fibres are used to make ropes. Kenaf matures in 100 to 200 days.

4.1.4 Structure of Kenaf

The fibres in kenaf are found in the bast (bark) and core (wood). The bast constitutes 40% of the plant. "Crude fibre" separated from the bast is multi-cellular, consisting of several individual cells stuck together. The individual fibre cells are about 2–6 mm long and slender. The cell wall is thick (6.3 μm). Kenaf fibre from bast could be gained as long as 2 meters and it becomes more widespread in polymer composite and concrete industry. The kenaf fibre needs to be treated properly to remove the lignin. The tensile strength of the kenaf fibre is about 800 MPa, which makes it suitable natural fibre in engineering applications. The core is about 60% of the plant and has fibre cells that are thick ($\approx 38 \mu\text{m}$) but short (0.5 mm) and thin-walled (3 μm).

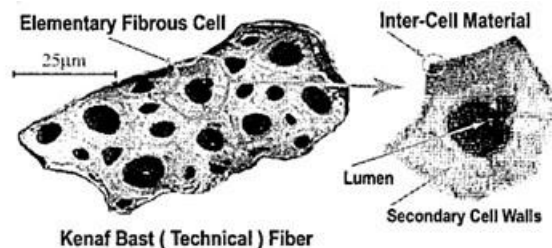


Figure 2: Structure of Kenaf

4.2 Methods

4.2.1 Procedure of Needle Punching

Fiber Preparation: Preparing the fibers used to make the fabric is the first stage in the production process for needle-punched nonwoven fabrics. This can entail separating the various fibre kinds into individual strands or combining them.

Web Formation: After that, the fibers are arranged into a web arrangement, the fibre pattern that will be utilized to make the fabric. Typically, a carding machine or other similar device creates this web shape by combing the fibers into the appropriate pattern.

Needle Punching: After the web formation is finished, the fibers are punched with needles to increase the strength of the fabric by interlocking the fibers together.

Finishing: Once the needle punching process is finished, the fabric undergoes a finishing process. This process can include the addition of a backing, like a scrim, or applying a coating to enhance the fabric's durability.

4.2.2 Sample Manufacturing Process

During the preparation process, two types of fibers are utilized: coir and kenaf. Coir is known for its dense structure, which can make it challenging to form a web during the manufacturing process. This can lead to complications such as needle breakages and increased time consumption. To address this issue, coir is treated with a 30% concentration of caustic soda for 24 hours, then washed and dried for further processing. To enhance the mat's water retention properties, kenaf fibers are blended with the coir fibers.

The non-woven mat is carefully crafted using a combination of 1.5 kilograms of coir and 1.5 kilograms of kenaf, with utmost precision and attention to detail. The kenaf fibre is expertly cut into 40mm lengths before the manufacturing process to ensure seamless blending with the coir and a smooth production process.

The fibers are then processed using a carding machine, a sophisticated tool that individualizes and aligns the fibers to form a web. The web is then subjected to a needle punching process with a density of 250 punches per inch, a technique that ensures the mat's durability and strength. The variables are mentioned in Table 1.

Overall, the production process involves the use of high-quality raw materials, with the total weight of the materials used amounting to approximately 3 kilograms. The result is a non-woven mat that is visually appealing but also resilient and long-lasting.

Table 1: The blend ratio variables in the non- woven mat manufacturing process

S. No.	Non -woven Blends (Coir/Kenaf)	Needle Punching (density/inch)	No. of Variables
1	50/50	250	1
2	75/25	250	1
3	25/75	250	1

5. Testing of Samples

- Water absorption (ISO 20158:2018)
- Water Retention (IHTM:55)
- Porosity (IHTM:77)

5.1 Water Absorption and Water Retention Procedure

Cut or prepare the sample material into standardized shapes or sizes, ensuring they are clean and free from any contaminants. Weigh the samples individually and record their initial dry weights. Submerge the prepared samples completely in water at a specified temperature (usually room temperature, unless otherwise specified). Ensure that the samples are fully submerged and not touching each other to prevent interference with water absorption. Start the timer or note the time when the immersion begins. Weigh each sample individually after removing surface water to determine its wet weight. Record the wet weight of each sample. Calculate the water absorption for each sample using the formula:

$$\text{Water absorption} = \frac{w_2 - w_1}{w_1} \times 100$$

Record the water absorption percentage for each sample

5.2 Porosity Procedure

Weigh the dry sample accurately using a balance. Record the initial weight (W1) of the sample. Determine the bulk volume of the sample by measuring its dimensions (length, width, and height) and calculating the volume. Alternatively, measure the volume displaced by immersing the sample in water in a measuring cylinder and recording the volume change. Submerge the sample completely in water or another suitable liquid. Allow the sample to saturate in the liquid for a specified duration. This could range from a few minutes to several hours depending on the material and testing requirements. Ensure the sample remains fully immersed during this period. Weigh the sample again after saturation, recording the saturated weight (W2). Calculate the porosity using the formula:

$$\text{Porosity \%} = \frac{W_2 - W_1}{\text{volume of sample}} \times 100$$

Higher porosity values may indicate greater void space within the material, which can affect properties such as strength, permeability, and thermal conductivity.

6. Result and Discussion

The experiment involved testing three different samples to evaluate their water retention ability and porosity level. Figure 3 shows the three different samples.

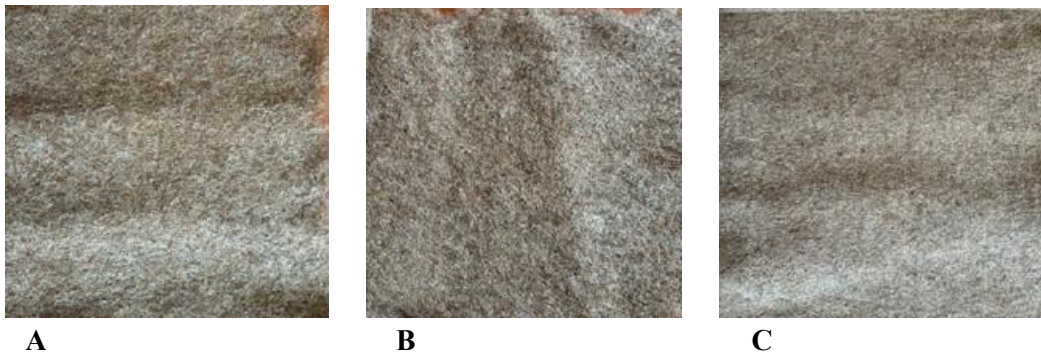


Figure 3: A) 50/50- Coir/Kenaf, B) 75/25- Coir/Kenaf, C) 25/75- Coir/Kenaf

6.1 Water Retention and Water Absorption

To ensure their water retention capabilities, the non-woven mats undergo rigorous testing. This involves submerging the sample mats in water for several minutes and measuring the results using a specific formula. This testing process allows us to determine the effectiveness of the mats and ensure that they will perform well in real-world situations.

$$\text{Water absorption} = \frac{w_2 - w_1}{w_1} \times 100$$

Table 3: Water Absorption and Retention Properties of Non-Woven Coir/Kenaf Mats with Different Composition Ratios

Non-woven mats (Coir/Kenaf)	Water absorption (minutes)	Water Retention %
50/50	6 minutes	97.4%
75/25	6 minutes 58 seconds	97.8%
25/75	6 minutes 45 seconds	97.2%

6.2 Porosity

Our non-woven mats undergo a rigorous testing process to evaluate their porosity properties. This includes a thorough pore size analysis to detect any possible water seepage and determine the retention capability of the mats. This guarantees that the mats meet the most stringent standards of quality and performance. Table 4 shows the pore size or porosity of the different non- woven mats.

Table 4: Porosity of non-woven mats with different compositions

S. No.	Non-Woven Mats (coir/Kenaf)	Pore Size %
1	50/50	0.60
2	75/25	0.50
3	25/75	0.48

6.3 Water absorption time and Porosity of the non-woven mats

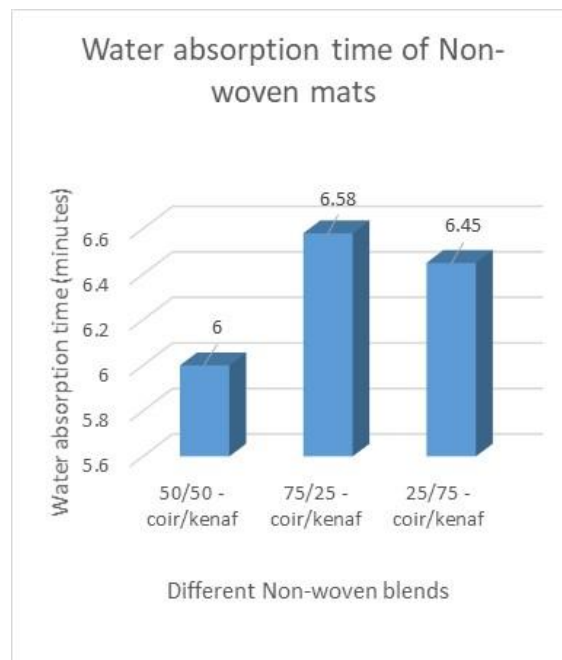


Figure 4: Water Absorption Time of the Non-woven Mats

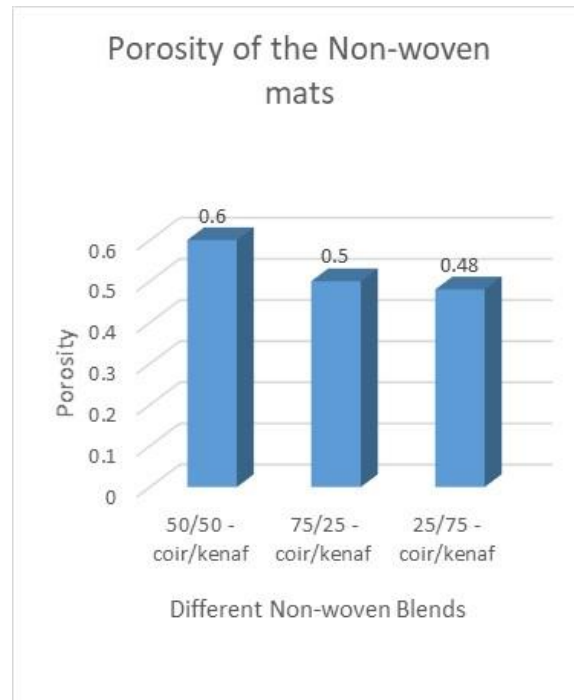


Figure 5: Porosity of the Non- woven Mats

The data presented in Figure 3 indicate the water absorption time for various non-woven mats. Notably, the 50/50 coir/kenaf mat, possessing a pore size of 0.6 (as depicted in Figure 4), absorbs water in a mere 6 minutes. On the other hand, the 75/25 coir/kenaf mat, having a pore size of 0.5 (also shown in Figure 4), absorbs water in 6 minutes and 58 seconds. The 25/75 coir/kenaf mat, with a pore size 0.48, takes 6 minutes and 45 seconds to absorb water. Although the pore size difference between the 75/25 and 25/75 coir/kenaf blended mats is marginal, the irregular blend ratio causes a discrepancy in water absorption time. It is worth noting that the 50/50 coir/kenaf blended mat displays a significantly larger pore size, resulting in faster water absorption compared to other samples.

6.4 Water Retention Percentage of the Non-woven mats

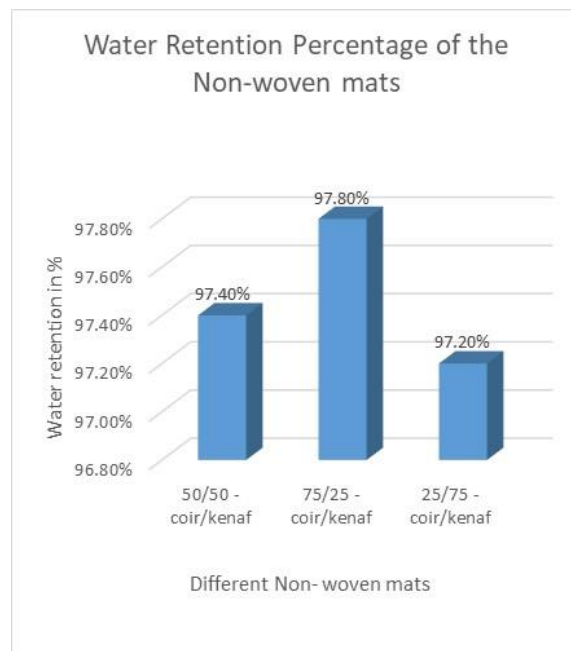


Figure 6: Water Retention % of Non-woven Mats

Figure 5 presents data on the water retention percentage of non-woven mats. This retention is directly related to the mats' porosity, as shown in Figure 4. Specifically, the 25/75 coir/kenaf blended mat has a much smaller pore size than the other non-woven mats, resulting in a higher water retention capacity that is water seepage is less after absorption.

7. Conclusion

Through careful experimentation and analysis of various non-woven mat samples incorporating different ratios of coir and kenaf fibers, we have discovered a remarkable finding. Our research indicates that Sample 3, which comprises 75% kenaf and 25% coir, exhibits superior water retention properties compared to other samples. With its dense and compact structure, coupled with a smaller pore size, it was able to retain a remarkable 97.2% of water within just 6 minutes and 45 seconds. This breakthrough discovery suggests that the blend of 75% kenaf and 25% coir is exceptionally effective in water retention, offering tremendous potential for addressing water scarcity issues in agriculture. With further optimization of this blend ratio and manufacturing process, we can achieve significant advancements in sustainable irrigation practices, thereby contributing to the conservation of water resources in agricultural settings.

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